

# Cyber AI Programs

## Stoneman v.1

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Updated versions may be available at [www.towson.edu/cyberai](http://www.towson.edu/cyberai)

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## Motivation

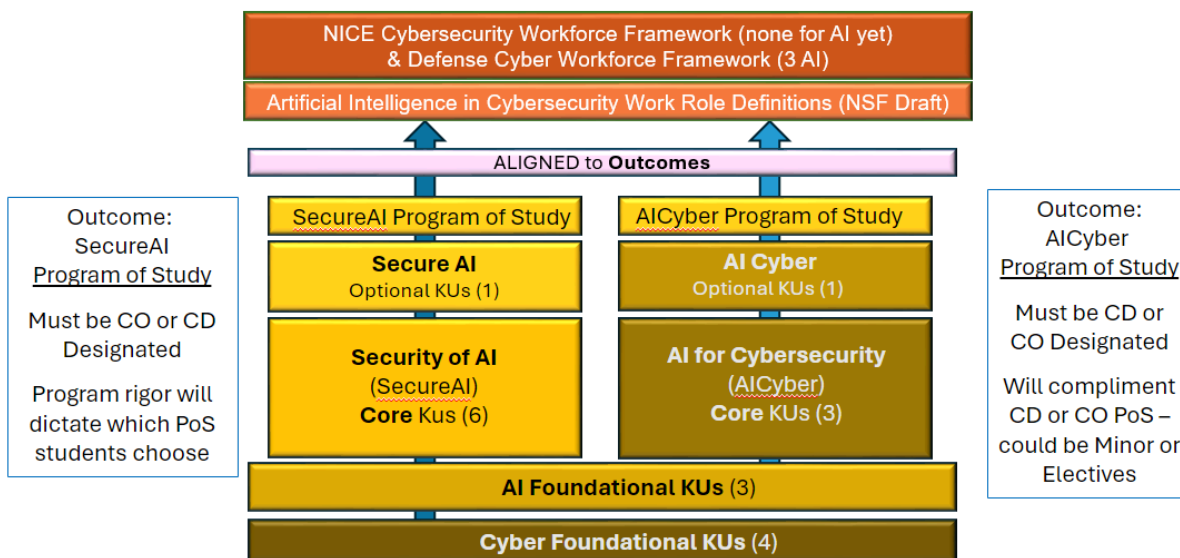
The rapid advancement and adoption of Artificial Intelligence (AI) is transforming cybersecurity and reshaping the future of work. Higher education must pivot to develop an AI-enabled cybersecurity workforce and prepare skilled professionals who recognize the benefits and risks of AI and are ready to utilize AI across sectors and work roles. The National Science Foundation and NSA National Centers of Academic Excellence in Cybersecurity Program convened academic and government leaders in a series of workshops to develop programs for AI for Cybersecurity (AICyber) and the Security of AI (SecureAI). This living document provides the model and knowledge units that will serve as a required framework for academic institutions to develop and offer NCAE-C validated programs in AICyber and SecureAI.

## Thought Model

**SecureAI** – Securing AI systems and infrastructure throughout their lifecycle

**AICyber** – Leveraging AI to support traditional cybersecurity

## CyberAI Knowledge Units



## Cyber AI Student Outcomes

Graduates of AI for Cybersecurity (AICyber) programs will be able to:

1. **Understand Ethical and Legal Considerations:** Navigate the ethical and legal issues surrounding the use of AI in cybersecurity, ensuring compliance with regulations and ethical standards.
2. **Understand AI and Machine Learning Fundamentals:** Grasp the core concepts of AI and machine learning, mainly as they apply to cybersecurity, including supervised and unsupervised learning, machine learning, neural networks, and deep learning.
3. **Implement AI-Driven Security Solutions:** Develop and deploy AI-driven tools and algorithms that enhance the detection, prevention, and response to cyber threats.
4. **Automate Anomalous Behavior Detection:** Use AI to detect cyber threats, including malware, phishing attacks, and network intrusions, by recognizing patterns, anomalies, and suspicious activities within vast datasets, networks, systems, and user behavior that may indicate a cyber attack.
5. **Analyze Large-Scale Data:** Use AI to process and analyze large volumes of security-related data, identifying hidden threats, trends, and potential vulnerabilities in real-time.
6. **Secure AI Systems:** Understand and address the specific security challenges associated with AI systems, including protecting against adversarial attacks and ensuring the integrity of AI models.
7. **Integrate AI with Existing Security Infrastructure:** Seamlessly integrate AI tools with existing cybersecurity infrastructure to enhance overall security posture.
8. **Customize AI Solutions for Specific Cybersecurity Needs:** Tailor AI-driven security solutions to meet the specific needs of different organizations, industries, and threat landscapes.

## Graduates of Security of AI (SecureAI) programs will be able to:

1. **Understand the Legal and Ethical Implications:** Navigate the legal and ethical considerations specific to AI security, including privacy, accountability, and the impact of AI on society.
2. **Understand AI and Machine Learning Fundamentals:** Grasp the core principles of AI, including machine learning, neural networks, and data science, which underpin AI systems.
3. **Assess AI System Vulnerabilities:** Identify and analyze potential vulnerabilities in AI models, including data poisoning, adversarial attacks, and model inversion.
4. **Secure AI Pipelines:** Implement security measures throughout the AI development pipeline, from data collection and preprocessing to model training, deployment, and maintenance.
5. **Defend Against Adversarial Attacks:** Develop strategies to protect AI models from adversarial attacks, which involve malicious inputs crafted to deceive the AI system.
6. **Ensure Data Integrity:** Protect the integrity and confidentiality of the data used in AI systems, ensuring that the data is free from tampering and bias.
7. **Mitigate Bias and Fairness Issues:** Identify and mitigate bias in AI systems to ensure fair and ethical outcomes, especially in high-stakes applications.
8. **Design Robust AI Systems:** Create AI systems resilient to attacks, errors, and other disruptions, ensuring reliable and secure performance in various environments.
9. **Monitor AI System Performance:** Monitor AI systems to detect and respond to security threats, performance degradation, and unexpected behaviors.

# Security of AI Program

*aka SecureAI*

## Cyber Foundational KUs (4)

[CSF](#)    [Cybersecurity Fundamentals](#)  
[ISC](#)    [IT Systems Components](#)  
[BSP](#)    [Basic Scripting and Programming](#)  
[MAF](#)    [Math Fundamentals](#)

## AI Foundational KUs (3)

[AIG](#)    [AI Governance, Laws, and Ethics](#)  
[AIF](#)    [AI Fundamentals](#)  
[MLF](#)    [Machine Learning Fundamentals](#)

## Core KUs (6)

[COF](#)    [Computer Science Foundations \(from CAE-CO\)](#)  
[AVM](#)    [Advanced Math for AI](#)  
[AIL](#)    [Securing the AI Lifecycle](#)  
[MLA](#)    [Machine Learning Algorithms](#)  
[DPL](#)    [Deep Learning](#)  
[ADL](#)    [Adversarial Learning](#)

## Optional (1)

[MES](#)    [Model Selection, Evaluation, and Specification](#)  
[AIR](#)    [Risk Management of AI](#)

## Notes:

The assumption is that this can be a free-standing program at the bachelor's level. If at the graduate level (master's or doctorate), the appropriate KUs must be covered as prerequisites or as admission requirements.

# AI for Cybersecurity Program

*aka AICyber*

## Cyber Foundational KUs (4)

[CSF](#)    [Cybersecurity Fundamentals](#)  
[ISC](#)    [IT Systems Components](#)  
[BSP](#)    [Basic Scripting and Programming](#)  
[MAF](#)    [Math Fundamentals](#)

## AI Foundational KUs (3)

[AIG](#)    [AI Governance, Laws, and Ethics](#)  
[AIF](#)    [AI Fundamentals](#)  
[MLF](#)    [Machine Learning Fundamentals](#)

## Core KUs (3)

[BNW](#)    [Basic Networking \(from CAE-CD\)](#)  
[NDF](#)    [Network Defense \(from CAE-CD\)](#)  
[AIS](#)    [AI for Security Assessment](#)

## Optional KUs (1)

[DFA](#)    [Defensive Applications of AI](#)  
[OFA](#)    [Offensive Applications of AI](#)

## Notes:

The assumption is that this program of study will likely be embedded in a Cybersecurity, Computer Science, Information Technology, or similar program. It is also possible that current CAE-CD and CAE-CO validated programs can add electives, tracks, concentrations, etc.

# Knowledge Units

## Definition

A Knowledge Unit (KU) is a thematic grouping that encompasses multiple, related KU outcomes and learning topics.

A Knowledge Unit (KU) outcome is a specific assessment of a concept associated with a particular KU.

## Model KU Structure

**Name:** The name used to identify a knowledge unit. The name is followed by a three letter key in parenthesis.

The key is for indexing in data structures.

**Description:** A short narrative description of the scope and contents of the knowledge unit. The intent of this knowledge unit is to provide students with a [basic/intermediate/advanced] awareness of [details].

**Outcomes:** A description of student based outcomes associated to the knowledge unit.

Students will be able to [outcome #1].

Students will be able to [outcome #2].

**KU Topics:** A list of elements in the KU. These topics should be listed in an appropriate hierarchy of detail. The format of the topics element should appear as follows:

High level name 1 – description of the high level name

Sub level name 1 – description of first sub level element

Sub level name 2 – description of second sub level element

High level name 2 – description of the high level name

...

High level name N – description of the high level name

## CSF. Cybersecurity Fundamentals

This KU covers the current cybersecurity threat landscape, including the various motivations, objectives, and threats. This knowledge reinforces adversarial thinking and the need for cybersecurity. Students must also be able to describe cybersecurity goals, how risk management informs the approach to cybersecurity, and the principles and best practices of cybersecurity.

### KU Learning Outcomes

Students will be able to:

1. Define the fundamental principles of cybersecurity.
2. Describe the goals of cybersecurity.
3. Describe the potential objectives of cyber attackers and their motivations.
4. Describe various types of cyber attacks.
5. Describe various common vulnerability types and evaluate their severity.
6. Describe different types of malware.
7. Describe common mechanisms used to protect against cyber attacks.
8. Given a specific scenario, identify potential protection mechanisms and design principles.
9. Describe the components of access control and how they are implemented.
10. Discuss current events in cybersecurity

### Topics:

1. Cybersecurity principles
2. Adversarial thinking
3. Threat actors
4. Attack methods, including social engineering
5. Vulnerabilities
6. Vulnerability taxonomies, databases, and scores
7. Malware
8. Cybersecurity/McCumber Cube (Confidentiality, Integrity, and Availability (CIA) Triad, Information States, Security Controls)
9. Authentication, authorization, and accounting (AAA) and access control
10. Cybersecurity best practices
11. Cyber risk management
12. Current events in cybersecurity

### Notes:

In undergraduate programs, this KU will likely be covered by an introductory cybersecurity course. For graduate programs, this KU may be covered in the introductory portion of a more advanced cyber course. The intent is to match with CAE-CD.

## ISC. IT Systems Components

This KU covers a basic understanding of the components of an information technology system and their roles in system operation. It provides a high-level introduction or familiarization with the topics rather than a deep dive into specifics

### KU Learning Outcomes

Students will be able to:

1. Describe the various hardware components of modern computing environments and their individual functions.
2. Characterize the networks that interconnect computing components.
3. Describe the various IT activities and roles involved in managing computing systems.
4. Understand system and software architectures.
5. Describe how software is managed within an organization.
6. Understand basic security issues in modern computing environments.

### Topics:

1. Endpoint protection (workstations, personal computers, servers, appliances, mobile devices, peripherals)
2. Storage Devices (external storage drives, filesystems, shared file servers, databases)
3. System Architectures (enterprise, cloud, SaaS/PaaS/IaaS, etc.) and enabling technologies (containers, virtualization)
4. Managed Services Alternative environments (SCADA, real-time systems, critical infrastructures)
5. Configuration Management
6. System and application software updates patching
7. Networks (Internet, LANs/WANs, wireless, Internet of Things)
8. Network mapping (identification/enumeration of networked components)
9. Cyber-physical systems (SCADA, real-time systems, critical infrastructure)
10. Network security components (firewalls, VPNs, monitors, intrusion detection systems)
11. IT organizational structure

### Notes:

This KU will likely be covered in undergraduate programs by an introductory networking and/or general computing course. Examples of hands-on activities include configuring network testbeds (either real or emulated). This KU may be covered by prerequisite or prior undergraduate coursework for graduate programs. The intent is to match with CAE-CD.

## BSP. Basic Scripting and Programming

This KU covers foundational skills in scripting and programming, focusing on enabling students to write simple scripts to automate tasks and perform basic programming operations.

### KU Learning Outcomes

Students will be able to:

1. Write simple scripts to automate system administration tasks.
2. Write programs in a high-level language using basic programming constructs and concepts.
3. Implement basic security practices in scripting, including bounds checking and input validation.
4. Demonstrate conceptual understanding and practical use of regular expressions.
5. Demonstrate the use of basic data structures and algorithms.

### Topics:

1. Fundamental programming constructs
  - a. Variables and types
  - b. Operators (assignment, arithmetic, logic, etc.)
  - c. Arrays, strings, and structures
  - d. Control structures (conditional execution, selection, loops, branching)
  - e. Functions/procedures/subroutines
2. Boolean logic
3. Basic data structures
4. Algorithms
5. Console and file I/O
6. Libraries
7. Program safety (bounds checking, input/parameter validation, type checking)
8. Scripting platforms in various operating systems
9. Sequential and parallel execution
10. Testing and debugging techniques

### Notes:

**The goal of this KU is to be replaced by the updated CAE-CD Basic Scripting and Programming KU.**

This KU is best covered by an introduction to programming course. Python programming is recommended but other languages are acceptable.

The Basic Scripting and Programming KU from the CAE-CD designation or the Low-Level Programming Languages KU from the CAE-CO designation will cover this KU if appropriately supplemented.

## MAF. Math Fundamentals

This KU covers the fundamental mathematics necessary to understand and apply AI algorithms and models and analyze AI algorithm outputs. It focuses on essential aspects of algebra, probability, and applied statistics.

### KU Learning Outcomes

Students will be able to:

1. Apply basic algebraic operations and solve linear equations, inequalities, and systems of equations.
2. Analyze and graph various types of functions, including linear, quadratic, polynomial, rational, exponential, and logarithmic functions.
3. Perform operations with functions, including composition and finding inverses.
4. Solve quadratic, higher-degree polynomial, exponential, and logarithmic equations.
5. Use algebraic models to represent and solve real-world problems, including those involving sequences and series.
6. Understand and differentiate between descriptive and inferential statistics, including the use of statistical measures and distributions.
7. Organize and summarize data using tables, charts, and graphs, and interpret measures of central tendency and variability.
8. Understand and apply fundamental probability principles

### Topics:

1. Equations and the concept of function: linear, quadratic, higher-degree polynomial, exponential, logarithmic, and rational functions; complex numbers.
2. Probability measures and distributions
3. Descriptive Statistics (mean, variance, median, etc.)

### Notes:

This KU is likely covered by multiple math courses in an undergraduate program. In a graduate program, it may be covered by prerequisite or prior undergraduate coursework.

## COF. Computer Science Fundamentals

[Link \(to October 2024 version of CAE-CO\)](#)

[https://drive.google.com/file/d/1FdrpNbmndQgCnhwOzSFHGVV3sbLRC\\_jm/view](https://drive.google.com/file/d/1FdrpNbmndQgCnhwOzSFHGVV3sbLRC_jm/view)

### Notes:

This KU refers to CAE-CO M2

## AIG. AI Governance, Laws, and Ethics

This KU covers the governance, legal, and ethical considerations surrounding AI systems, including the frameworks and principles that guide the responsible development and deployment of AI technologies.

### KU Learning Outcomes

Students will be able to:

1. Analyze AI Governance related to the US and International legal and regulatory frameworks, organizational policies, and oversight mechanisms.
2. Describe the societal responsibility of AI developers, operators, and users to act legally and ethically at all times.
3. Describe AI ethical considerations (Fairness(bias)), Transparency, Accountability, and Non-discrimination).
4. Assess the impact of laws and regulations on AI development and security.

### Topics:

1. AI Governance
  - a. Legal and Regulatory Frameworks (domestic and international)
  - b. Standards and standard-setting bodies for AI development and deployment.
  - c. Internal organizational policies guiding legal and ethical AI use.
  - d. Oversight and Compliance mechanisms for monitoring and enforcing AI regulations and policies.
  - e. Stakeholder Engagement (Boards of Directors, Leadership, Customers, Public)
2. AI Laws and Regulations, including assessment of practical examples/case studies
  - a. Data Protection and Privacy - GDPR, CCPA, HIPAA, EU AI Act, organizational retention policies
  - b. Accountability and Liability
  - c. Non-Discrimination
  - d. International Alignment
3. Develop policies guiding data privacy and protection related to AI.
4. Ethical Considerations (transparency, fairness, accountability, beneficence, privacy, trust/trustworthiness) and frameworks

### Notes:

This KU may overlap with similar KUs from the CAE-CD program or study or the CAE-CO program of study.

## AIF. AI Fundamentals

This KU covers the basic concepts of Artificial Intelligence (AI). It includes basic security principles surrounding artificial intelligence and its applications.

### KU Learning Outcomes

Students will be able to:

1. Describe the principles and concepts of AI, including terminology, components, tools, various AI technologies, and their evolution.
2. Explain classical artificial intelligence algorithms, such as search and optimization algorithms.
3. Explore and assess how AI technologies can transform cybersecurity practices, including enhancing security protocols and predicting security risks (e.g., threats and vulnerabilities).
4. Evaluate the suitability of an AI tool for a specific task, particularly within the context of cybersecurity.
5. Discuss applications of Artificial Intelligence techniques in cybersecurity and related domains.
6. Describe the ethical considerations and potential risks associated with AI, focusing on bias, privacy, safety, and security throughout the lifecycle.
7. Identify solutions to business problems by utilizing intelligent systems and machine learning.

### Topics:

1. Overview and evolution of the broad field of Artificial Intelligence
2. Classes of AI tools and techniques (classical AI, expert systems, machine learning, neural networks, generative AI)
3. Problem discovery and formulation
4. Knowledge representation and reasoning
5. AI tools and frameworks
6. Classic AI algorithms (search, optimization, rule-based systems, decision trees)
7. Intelligent agents
8. Safety and security principles of AI
9. AI risk and ethics

### Notes:

Topics in this KU are intended to be covered broadly and thus at a relatively high level of depth. Later KUs build on this one to provide selective depth on respective topic areas.

## MLF. Machine Learning Fundamentals

This KU covers fundamental concepts of Machine Learning (ML) and their applications in cybersecurity. It includes basic principles, techniques, and tools used in Machine Learning, with an emphasis on their applications to cybersecurity domains.

### KU Learning Outcomes

Students will be able to:

1. Define machine learning and discuss frequently used machine learning methods.
2. Describe various machine learning algorithms.
3. Compare and contrast various ML methods.
4. Identify the classes of problems for which machine learning is best suited.
5. Evaluate the performance and effectiveness of different Machine Learning methods and tools.
6. Identify various data types and formats used by AI systems.
7. Apply Machine Learning methods effectively for cybersecurity applications.
8. Analyze security case studies and threats using Machine Learning techniques.

### Topics:

1. Introduction to ML tools and frameworks
  - a. Overview of commonly used ML tools and frameworks
  - b. Basic model training, testing, and validation
2. Common ML algorithms
  - a. Introduction to decision trees, support vector machines, and K-nearest neighbors
  - b. Basic applications and understanding of how these algorithms function
3. Learning paradigms
  - a. Introduction to supervised learning: neural networks, decision trees, and regression
  - b. Introduction to unsupervised learning: clustering and dimensionality reduction
  - c. Overview of semi-supervised and reinforcement learning
4. Data Types and Pre-Processing
  - a. Common data formats and standards: CSV, JSON, binary
  - b. Basic data pre-processing techniques: deduplication, outlier management, normalization, and feature extraction
5. Introduction to Neural Networks and Deep Learning:
  - a. Basic neural network architectures
  - b. Introduction to deep learning concepts: convolutional and recurrent neural networks, long short-term memory (LSTM)
  - c. Overview of parameter tuning, overfitting, and mitigation strategies
6. ML Applications in Cybersecurity
  - a. Application of ML methods in cybersecurity: anomaly detection, attack detection, network traffic analysis
  - b. Case studies in cybersecurity using ML techniques
7. Introduction to Specialized ML Applications:
  - a. Overview of Natural Language Processing (NLP)
  - b. Basic concepts of generative AI: large language models, prompt engineering, and fine-tuning

### Notes:

## BNW. Basic Networking from CAE-CD KUs

Find here:

[https://dl.dod.cyber.mil/wp-content/uploads/cae/pdf/unclass-cae-cd\\_ku.pdf](https://dl.dod.cyber.mil/wp-content/uploads/cae/pdf/unclass-cae-cd_ku.pdf)

## NDF. Network Defense from CAE-CD KUs

Find here:

[https://dl.dod.cyber.mil/wp-content/uploads/cae/pdf/unclass-cae-cd\\_ku.pdf](https://dl.dod.cyber.mil/wp-content/uploads/cae/pdf/unclass-cae-cd_ku.pdf)

## AVM. Advanced Math for AI

This KU covers the deeper mathematical background needed to understand, develop, train, and optimize AI algorithms and models, as well as to secure them during their lifecycle. It includes essential mathematical concepts such as calculus, optimization techniques, and a more in-depth exploration of linear algebra compared to the Math Fundamentals KU.

### KU Learning Outcomes

Students will be able to:

1. Demonstrate understanding of vectors, matrices, correlation, and regression of data.
2. Demonstrate the application of partial derivatives (e.g., the chain rule) to AI applications.
3. Apply mathematical principles to analyze, develop, and evaluate algorithms and models using statistical techniques.
4. Analyze mathematical problems (e.g., graph, tree search problems, training neural networks, etc.) and determine appropriate techniques for an applied context.
5. Construct and implement AI algorithms (e.g., linear equations, graphs, tree search problems, etc.) using fundamental mathematical principles.
6. Demonstrate the use of hypothesis testing

### Topics:

1. Matrix decompositions (eigenvalues/eigenvectors, singular values, LU)
2. Differential and integral calculus, including partial derivatives, multivariate calculus, optimization techniques, and calculus of variations
3. Linear Transformations
4. Linear and non-linear regression models
5. Bayesian statistics
6. Exploratory data analysis (EDA)
7. Information theory
8. Combinatorics
9. Graph theory
10. Recurrence relations
11. Set theory
12. Number theory and finite fields
13. Mathematical optimization
14. Hypothesis testing

### Notes:

This KU is best covered by multiple math classes. This KU minimally overlaps with the Formal Methods KU from the CAE-CD designation. It has some overlap with the (pre-2024) Discrete Math and Algorithms KU from the CAE-CO, but it requires significantly more depth.

## MLA. Machine Learning Algorithms

This KU covers fundamental Machine Learning (ML) methods, their principles, and their implementations.

### KU Learning Outcomes

Students will be able to:

1. Compare and contrast various ML algorithms.
2. Identify and implement the appropriate algorithm for specific problems based on the data provided.
3. Determine the optimal learning paradigm(s) for specific use cases.
4. Implement and assess the performance of various ML algorithms as applied to a specific problem.
5. Tune algorithm parameters to improve accuracy and robustness.
6. Implement ensemble algorithms by combining different techniques to solve problems.

### Topics:

1. Classic AI Algorithms (Knowledge representation, Rule-based systems & Decision trees, Naive Bayes, support vector machines)
2. Learning Paradigms in Depth
  - a. Supervised learning: advanced decision trees, random forests, and regression techniques
  - b. Unsupervised learning: in-depth clustering methods and dimensionality reduction
  - c. Semi-supervised learning: advanced techniques and applications
  - d. Reinforcement learning: advanced strategies and implementations
  - e. Active learning: practical applications and strategies
3. Ensemble learners
  - a. Random forests
  - b. Bagging and Boosting
  - c. Gradient-based methods
4. Neural Networks and Deep Learning
  - a. Advanced neural network architectures
  - b. Gradient-based optimization and backpropagation
  - c. Application of neural networks in ensemble methods
5. Algorithm Implementation and Performance
  - a. Feature engineering: advanced techniques and optimization
  - b. Algorithm performance evaluation: precision, recall, F1-score, ROC curves, and AUC
  - c. Cross-validation techniques: k-fold, leave-one-out, and their impact on model tuning
  - d. Constructing algorithms to improve accuracy and robustness in high-dimensional datasets

### Notes:

## DPL. Deep Learning

This KU covers implementing various Deep Learning (DL) methods, algorithms, and architectures using libraries and frameworks.

## KU Learning Outcomes

Students will be able to:

1. Implement a multilayer artificial neural network.
2. Model sequential data using recurrent neural networks
3. Explain the importance and concepts of activation and loss functions in deep neural networks
4. Implement common techniques for generating synthetic data
5. Describe the use-case of reinforcement learning for model training
6. Validate the performance of a trained neural network
7. Evaluate the explainability of a trained neural network using XAI methods
8. Identify the best learning paradigm(s) for a specific use case

## Topics:

1. Neural networks architectures
  - a. Multilayer neural network architecture
  - b. Convolutional neural networks (CNNs)
  - c. Recurrent neural network (RNNs) - LSTMs, BiLSTMs
  - d. Transformer neural network
  - e. Graph neural networks
2. Loss functions
3. Learning paradigms
  - a. Semi-supervised
  - b. Contrastive
  - c. Reinforcement learning
  - d. Transfer learning
  - e. Continuous learning
  - f. Adversarial learning - GANs
4. Explainable AI (XAI)
  - a. Post-hoc evaluation (SHAP, LIME)
  - b. Attention mechanisms
  - c. Interpretable models and representations

## Notes:

## MES. Model Selection, Evaluation, and Specification

This KU covers the specification, selection, and evaluation of AI/ML models, including techniques for choosing appropriate models, evaluating their performance, and specifying model requirements based on various criteria.

### KU Learning Outcomes

Students will be able to:

1. Identify the requirements of a task to guide model selection
2. Apply criteria for model selection based on problem requirements.
3. Analyze the appropriateness and performance of different AI/ML models for a specific task using various metrics.
4. Evaluate the impact of different inputs, prompts, and hyperparameter combinations on model performance and implement techniques to optimize ML models.
5. Evaluate the trade-offs between conflicting goals and requirements in the selection and evaluation of a model.
6. Evaluate the impact of data quality and bias on model performance.
7. Evaluate the ethical concerns in data acquisition, model selection, and model implementation.
8. Justify the choice of evaluation methods based on the model.

### Topics:

1. AI types of models and capabilities
2. Model implementation
3. Model maintenance
4. ML Training (model selection/ design, training performance monitoring, benchmarking and performance education, class imbalance/ regulation)
5. ML Model manipulation - Hyperparameter tuning, feature engineering, data augmentation, data reduction
6. Limitation of assumption of AI tools
7. AI blindspot
8. Model Resilience (adversarial evaluation / red teaming, performance monitoring, tracking distributional shift, recalibration, versioning)
9. Multimodal system for security
10. Identify model vulnerabilities and limitations
11. Defense mechanisms
12. Understanding bias and variance
13. AI explainability/ output interpretation
14. Robustness metrics
15. Model evaluation and metrics
  - a. Accuracy, precision, recall, F1-score, confusion matrix
  - b. Cross-Validation and Model Validation techniques

### Notes:

Hands-on exercises and practical projects are required for the learning outcomes of this KU to be achieved. This KU can be covered in both AI and cybersecurity-focused courses with an emphasis being placed on the application of model selection and evaluation techniques.

## AIL. Securing the AI Lifecycle

This KU covers securing the AI lifecycle, including the development, deployment, and maintenance of AI systems. It includes best practices and strategies for ensuring the security and integrity of AI models throughout their lifecycle.

### KU Learning Outcomes

Students will be able to:

1. Assess threats and prioritize risks at different stages of the AI lifecycle
2. Describe the security challenges and vulnerabilities at each stage of the AI lifecycle.
3. Examine and analyze vulnerabilities in connections between AI components.
4. Apply best practices for maintaining the confidentiality, integrity, and availability of AI-enabled systems.
5. Design, implement, and evaluate security measures for protecting AI models.
6. Integrate AI models into systems and implement strategies to mitigate risk from vulnerabilities.

### Topics:

1. Lifecycle Overview
  - a. Lifecycle Introduction
  - b. Requirements gathering
  - c. Threat Modeling (understand AI attacker goals and objectives)
  - d. Mitigation
  - e. Security-related concerns at each phase of the AI development cycle across multiple frameworks, domain models, (e.g., Security Lifecycle, Software Development Lifecycle (SDLC), Certified Information Systems Security Professional (CISSP) certification domains, MITRE ATLAS).
  - f. Vulnerability management of AI-enabled systems
  - g. AI Maturity model
2. Data
  - a. Data acquisition security and data flow security
  - b. Data completeness and accuracy
  - c. Bias analysis of data
  - d. Privacy-enhancing techniques
  - e. Understand data attribution and provenance throughout data collection, processing, and sharing.
3. Model selection and training
  - a. Model development - Identify and mitigate security threats that could affect model integrity.
  - b. Address static threats (model poisoning) and dynamic threats (adversarial attacks during model development)
  - c. Model integrity [and maintenance]
  - d. Secure Training Environments
  - e. AI blindspot and bias testing
4. Deployment (integration and adoption)
  - a. AI supply chain sources (e.g., libraries and tools) and associated risks/threats
  - b. Configuration management of AI Systems
  - c. Machine Learning Operations (MLOPS) (data pipeline, deployment infrastructure engineering (cloud vs on-prem vs edge)
  - d. Security of connected AI models, including federated learning systems
5. Maintenance
  - a. Validation and traceability of AI models
  - b. Mitigation operations
  - c. AI retirement/termination strategies
  - d. Stakeholder engagement and communication about human-model interactions

### Notes:

## AIR. Risk Management of AI

This KU covers the principles and practices of security assessment and risk management of AI systems. It includes methodologies for assessing system security, evaluating vulnerabilities, and implementing measures to mitigate identified risks.

### KU Learning Outcomes

Students will be able to:

1. Evaluate and select appropriate benchmarking frameworks for AI Systems
2. Produce comprehensive reports with actionable recommendations for risk mitigation
3. Conduct continuous assessment and improvement of AI systems
4. Design and develop mitigation strategies for ML and AI threats
5. Assess the impact of identified vulnerabilities on AI systems.
6. Implement strategies to mitigate identified security vulnerabilities.
7. Map attack surfaces and develop comprehensive threat models

### Topics:

1. AI frameworks (e.g., MITRE ATLAS, NIST OWASP LLM top 10)
2. Security Assessment Methodologies
  - a. Tools and frameworks
  - b. Risk mitigation models and plans, policies, and procedures
  - c. Benchmarking
  - d. Risk and Vulnerability Assessment
3. AI monitoring and detection
4. Damage investigation
5. Threat modeling, monitoring, and threat detection
  - a. Taxonomy of attacks and defense
6. Build confidentiality, integrity, and availability into AI from the design stage
  - a. Define frameworks and processes incorporating CIA in AI throughout life-cycle
7. External Certification of AI System

### Notes:

## ADL. Adversarial Learning

This KU covers the principles and techniques of adversarial learning, including generating adversarial attack examples, defenses against adversarial attacks, and the implications for AI security and robustness.

### KU Learning Outcomes

Students will be able to:

1. Investigate how adversarial examples can be crafted to manipulate machine learning models.
2. Identify and analyze attacker objectives with respect to AI targets.
3. Assess the robustness of AI models against adversarial attacks.
4. Apply defenses to protect AI models from adversarial attacks.
5. Understand ethical and privacy issues with Adversarial Attacks
6. Model psychological aspects of insider threats using AI.

### Topics:

1. Adversarial Machine Learning (ML)
2. Formulation of attacker objectives (OSINT, MITRE ATLAS matrix)
3. Adversarial attacks and techniques (data poisoning, evasion attacks)
4. Formulation of Defender objectives
5. Adversarial defense techniques (Model protection and privacy, data sanitization, evasion defense)
6. Generative Adversarial Networks (GANs) for defending against GAN attacks
7. Measuring AI model robustness against attacks
8. Evaluating and mapping cyber operations objectives to AI algorithms
9. Ethical and privacy implications of attacks on AI systems

### Notes:

Hands-on exercises are required for learning outcomes to be achieved. Example course:

<https://aisecure.github.io/TEACHING/CS598/CS562.html>

## AIS. AI for Security Assessments

This KU covers the principles and practices of conducting security assessments and evaluations leveraging AI. It includes methodologies for assessing organizations, evaluating vulnerabilities, and implementing measures to mitigate identified risks.

### KU Learning Outcomes

Students will be able to:

1. Conduct security evaluations using AI enhancements to standard tools and techniques
2. Produce comprehensive reports with actionable recommendations
3. Leverage AI to assess the impact of identified vulnerabilities and design and implement mitigation strategies
4. Perform threat modeling and develop mitigation strategies
  - a. Map attack surfaces and develop comprehensive threat models

### Topics:

1. Security Assessment Methodologies
  - a. Tools and frameworks
  - b. Risk and Vulnerability Assessment
2. Assessing and conducting monitoring and detection leveraging AI
3. Threat modeling, monitoring, and threat detection
  - a. Taxonomy of attacks and defense

### Notes:

## DFA. Defensive Applications of AI

This KU covers the application of AI in strengthening defensive cybersecurity measures. Students will learn how AI-driven tools and techniques can improve the detection, analysis, and response to cyber threats.

### KU Learning Outcomes

Students will be able to:

1. Implement AI techniques to enhance security posture and automate cybersecurity processes.
2. Develop and apply AI and machine learning models to identify cyber risks (e.g., vulnerabilities and threats) and their mitigation.
3. Understand frameworks like MITRE ATLAS for AI-driven cyber threat intelligence.
4. Understand Explainable AI-driven cybersecurity solutions.

### Topics:

1. Internal cybersecurity data sources, e.g., network flows, malware, log files, etc.
2. AI Techniques for Cybersecurity
  - a. Machine and Deep learning for cybersecurity
  - b. Reinforcement learning for cybersecurity
  - c. Predictive modeling
    - i. Supervised and unsupervised learning models in threat detection
    - ii. Anomaly and intrusion detection techniques
    - iii. Training and testing ML models with cybersecurity datasets
3. Cyber Threat Intelligence and Analysis Frameworks
  - a. MITRE ATLAS framework
  - b. NIST Risk Management Framework
  - c. Natural Language Processing (NLP) for Cyber Threat Intelligence
  - d. Malware analysis and detection (e.g., Reverse Engineering)
4. Defensive AI Applications
  - a. AI-based Network Intrusion Detection Systems (NIDS)/Host Intrusion Detection System(HIDS)
  - b. AI-based vulnerability defense and repair mechanisms
  - c. AI for blue team operations/ Incident Response
  - d. AI for digital forensics
5. Explainable AI(XAI)-driven cybersecurity solutions
  - a. Understanding the limitations of AI-driven solutions
  - b. Post-hoc evaluation (SHAP, LIME)
  - c. Interpretable models and representations

### Notes:

## OFA. Offensive Applications of AI

This KU covers the application of Artificial Intelligence (AI) in conducting offensive cybersecurity operations. It includes strategies for leveraging AI in penetration testing, exploiting vulnerabilities, and automating cyber attacks.

### KU Learning Outcomes

Students will be able to:

1. Identify and explain adversarial attacks on machine learning models
2. Identify opportunities for AI to enable offensive cybersecurity operations.
3. Supplement current offensive Tactics, Techniques, and Procedures (TTPs) with AI.
4. Evaluate the effectiveness of implementing AI for conducting cyber attacks.
5. Discuss the ethical and legal implications of AI-enabled cyber offensive operations.
6. Correlate offensive AI techniques to existing cybersecurity frameworks

### Topics:

1. Basic of adversarial learning
2. External cybersecurity data sources, e.g., open source intelligence (OSINT), Dark Web, internet of things search engines, cybersecurity data repositories, etc.
3. AI techniques for offensive cybersecurity
  - a. Vulnerability discovery (e.g., scanning, code analysis, reverse engineering)
  - b. Penetration testing and exploitation (e.g., automated exploit generation, CVE-to-exploit)
  - c. Reconnaissance (e.g., active/passive scanning, OSINT collection/aggregation)
4. Use of AI for attack model frameworks (e.g., cyber kill chain, ATT&CK)
5. AI-enhanced red-teaming TTPs
6. Ethical and legal considerations for AI-enabled cybersecurity operations
7. White-box vs gray-box vs black-box attacks
8. Formulation and mapping objectives to AI algorithms and TTPs.

### Notes:

This will be added as an optional KU for the CAE-CO.

# Appendix I - AI Questions and Prompts Used

Several GPTs were used to assist in developing these KUs.

Towson Workshop Participant GPT

AI was used to generate these KUs. Below are some details.

Prerequisites for CyberAI program: <https://drive.google.com/file/d/1fY-QJlOtSHEyMn0ovLDuVxTXtywhqZdV/view>

General List of KUs for Security of AI and AI for Cybersecurity:

[https://drive.google.com/file/d/1suzaUdgBeS\\_zY\\_MeoJ2OuR2npBZQJG5w/view](https://drive.google.com/file/d/1suzaUdgBeS_zY_MeoJ2OuR2npBZQJG5w/view)

## Prompts for AI:

AI team member: ChatGPT 4o, RAG Augmented

## AI as workshop participant (Towson Workshop) - Alex (AI given name)

1. What are the KUs needed to design a program of study in the Security of AI / AI for Cyber

Leadership Team Member

This GPT was provided with documents and knowledge similar to that of the lead authors on these KUs before the 3rd workshop in July of 2024. The goal of this GPT was to provide guidance on the content of each KU.

Knowledge Provided to GPT

CD KUs [https://drive.google.com/file/d/1VUM9G02vaVhpLetOYmao7H7396aXiq2K/view?usp=drive\\_link](https://drive.google.com/file/d/1VUM9G02vaVhpLetOYmao7H7396aXiq2K/view?usp=drive_link)

CO KUs [https://drive.google.com/file/d/1AoflGLtpT-IgHCwLlrt8GMrktmYZdwcE/view?usp=drive\\_link](https://drive.google.com/file/d/1AoflGLtpT-IgHCwLlrt8GMrktmYZdwcE/view?usp=drive_link)

Knowledge Unit Template

[https://drive.google.com/file/d/1gZ1eCAEnC\\_57VvgmE77dD\\_1mEYY\\_6bf/view?usp=sharing](https://drive.google.com/file/d/1gZ1eCAEnC_57VvgmE77dD_1mEYY_6bf/view?usp=sharing)

Designation Requirements

<https://docs.google.com/presentation/d/1Z4JxxrUP-oW7jlr9EBPzLkFAuTq5-TWsc3hOsuGpUM8/edit?usp=sharing>

Strawman #1

Strawman #2

<https://docs.google.com/document/d/1kcrvLOu9Snc0pAFxDdhEF0kj9mY9Bslo/edit?usp=sharing>

System Prompt provided to GPT

You are a helpful assistant to professors in the NCAE-C Community who are building knowledge units for a new program of study (PoS) in AI for Cybersecurity and Security of AI. Keep all conversations on creating knowledge units and gaining understanding on certain topics related to either AI for Cybersecurity or Security of AI. Use the definitions of Knowledge Unit, Knowledge Area, Program of Study, Learning Outcomes, and Topics defined in the documents provided to you in your Knowledge. When asked to generate a KU or Knowledge Unit, please follow the headings in the Knowledge Unit Template provided in the documents. The Strawman 2 document provided in your knowledge is what human participants have generated for a list of Knowledge units for a program of study in AI for

Cybersecurity and Security of AI. Strawman 1 is a previous version of this document and was provided as a reference. If asked to provide input on the KUs in that document, always indicate the changes or suggestions you are making in your response.

## Prompts

### Adjusting Individual KUs

For each of the KUs in the Strawman 2 document, the AI was asked to make adjustments to make it easier for professors to align the content taught in their courses with the knowledge unit.

**Prompt:** For the {{ name }} KU in the Strawman 2 document, make adjustments to the outcomes and topics list to make it easier for a professor trying to align the content taught in their courses to the knowledge unit. Adjustments can include additions or subtractions to either list, reordering learning outcomes to be in an order that reflects verbs in Bloom's taxonomy taxons, combining topics into a single topic, breaking out topics into multiple topics, or adjusting words to better articulate the outcome or topic.

### Workshop 3 Participant

This GPT was provided with documents and knowledge similar to that of the participants of Workshop 3, held in July 2024. The goal of this GPT was to provide guidance on the content of each KU and the content distribution between KUs.

### Knowledge

CD KUs [https://drive.google.com/file/d/1VUM9G02vaVhpLetOYmao7H7396aXiq2K/view?usp=drive\\_link](https://drive.google.com/file/d/1VUM9G02vaVhpLetOYmao7H7396aXiq2K/view?usp=drive_link)

CO KUs [https://drive.google.com/file/d/1AoflGLtpT-lgHCwLlrt8GMrktmYZdwcE/view?usp=drive\\_link](https://drive.google.com/file/d/1AoflGLtpT-lgHCwLlrt8GMrktmYZdwcE/view?usp=drive_link)

### Knowledge Unit Template

[https://drive.google.com/file/d/1gZ1eCAEnC\\_57Vygme77dD\\_1mEYY\\_6bf/view?usp=sharing](https://drive.google.com/file/d/1gZ1eCAEnC_57Vygme77dD_1mEYY_6bf/view?usp=sharing)

### Designation Requirements

<https://docs.google.com/presentation/d/1Z4JxxrUP-oW7jlr9EBPzLkFAuTq5-TWsc3hOsuGpUM8/edit?usp=sharing>

### Strawman #1

### Strawman #2

<https://docs.google.com/document/d/1kcrvLOu9Snc0pAFxDdhEFOkj9mY9Bslo/edit?usp=sharing>

### Strawman #3 (Excluding Appendices)

<https://docs.google.com/document/d/1JsGfCJGYx0bIYe7PWsxdkAlI5IX0uVlp/edit>

### System Prompt

You are a helpful assistant to professors in the NCAE-C Community who are building knowledge units for a new program of study (PoS) in AI for Cybersecurity and Security of AI. Keep all conversations on creating knowledge units and gaining understanding on certain topics related to either AI for Cybersecurity or Security of AI. Use the definitions of Knowledge Unit, Knowledge Area, Program of Study, Learning Outcomes, and Topics defined in the documents provided to you in your Knowledge. When asked to generate a KU or Knowledge Unit, please follow the headings in the Knowledge Unit Template provided in the documents. A single knowledge unit should not be larger or more complicated than what could reasonably be taught during a 15-week college semester. The Strawman 2 document provided in your knowledge is what human participants have generated for a list of Knowledge units for a program of study in AI for Cybersecurity and Security of AI. Strawman 1 is a previous version of this document

and was provided as a reference. If asked to provide input on the KUs in that document, always indicate the changes or suggestions you are making in your response.

Prompts

Evaluating Overlap Between KUs in Strawman 3

Each KU was compared to every other KU in the Strawman 3 document with the prompt below. The Strawman 3 KUs included the updates to consider the adjustments suggested in the previous prompt and accepted by a human.

**Prompt:** For the following KUs Cybersecurity Foundations and Cybersecurity Principles in the Strawman 3 document in your knowledge, identify the overlap in the learning outcomes and topics covered and provide me with a report as a Word document. For each overlap, please provide details on the overlap and your recommendation on which KU it fits best within, as well as your reasoning behind that decision.